

1 Solution — GIAM 2.5

Problem 1

1.1 Problem

There is a common variant of the existential quantifier, $\exists!$. Writing $\exists!x, P(x)$ asserts that there is a **unique** element in the universe making $P(x)$ true. Determine how to negate the sentence $\exists!x, P(x)$.

1.2 Answer

$$\neg[\exists!x, P(x)] \cong \underbrace{(\forall x, \neg P(x))}_{\text{nothing works}} \vee \underbrace{(\exists y \exists z, P(y) \wedge P(z) \wedge y \neq z)}_{\text{two different things work}}$$

In words: **either nothing satisfies P , or at least two different things do.**

An equivalent single-quantifier form, proved below, is

$$\neg[\exists!x, P(x)] \cong \forall x [P(x) \Rightarrow \exists y (P(y) \wedge y \neq x)],$$

read as "**every** witness has a companion" — no element is ever the lone one.

1.3 Why: Unpack “Exactly One” First

The key is that $\exists!$ is not a primitive — it abbreviates a **conjunction of two claims**:

$$\exists!x, P(x) \cong \underbrace{\exists x, P(x)}_{\text{existence}} \wedge \underbrace{\forall y \forall z [(P(y) \wedge P(z)) \Rightarrow y = z]}_{\text{uniqueness}}.$$

Since it is a conjunction, De Morgan says its negation is a **disjunction** — the claim can fail in exactly two ways, and negating must cover both:

$$\neg(\text{existence} \wedge \text{uniqueness}) \cong \neg\text{existence} \vee \neg\text{uniqueness}.$$

Negating each half in turn (flipping quantifiers as in this section):

$$\begin{aligned}
\neg \text{existence} &\cong \neg \exists x, P(x) \cong \forall x, \neg P(x), \\
\neg \text{uniqueness} &\cong \neg \forall y \forall z [(P(y) \wedge P(z)) \Rightarrow y = z] \\
&\cong \exists y \exists z \neg [(P(y) \wedge P(z)) \Rightarrow y = z] \\
&\cong \exists y \exists z [P(y) \wedge P(z) \wedge y \neq z].
\end{aligned}$$

The second line uses the negation of a conditional from Problem 2.2.4, $\neg(P \Rightarrow Q) \cong P \wedge \neg Q$. Joining the two halves with $\vee$ gives the boxed answer. ■

1.4 The Trichotomy

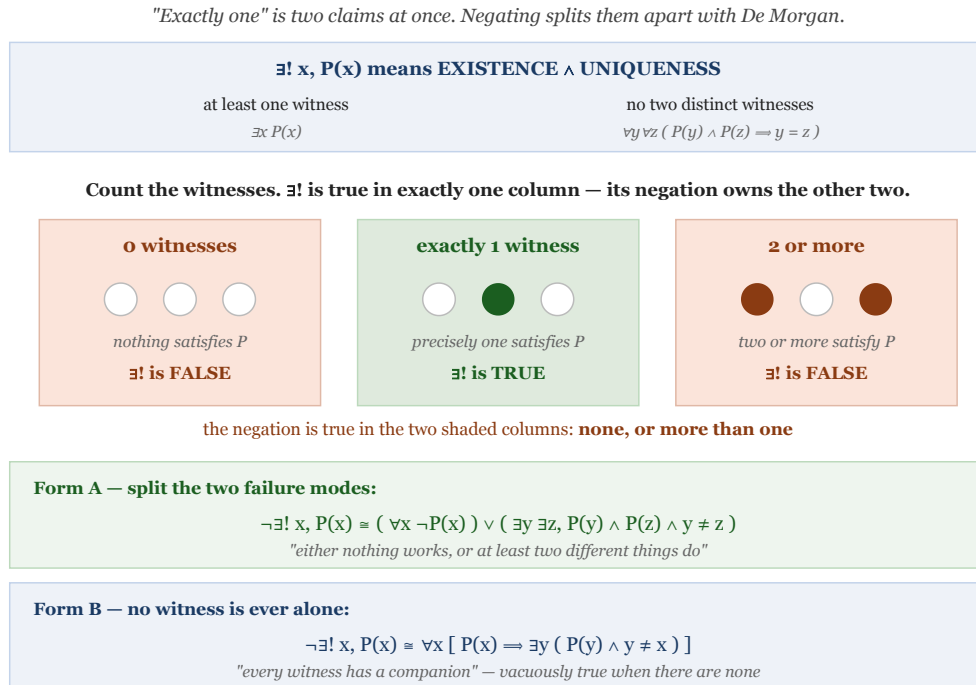


Figure 1.1: The unpacking: exists-unique $\exists! x P(x)$ means existence and uniqueness — at least one witness, written there exists $\exists x P(x)$, and no two distinct witnesses, written for all y and z , if $P(y)$ and $P(z)$ then y equals z . Then a trichotomy by witness count. Zero witnesses, shaded: nothing satisfies P , so exists-unique is false. Exactly one witness, green: precisely one satisfies P , exists-unique is true. Two or more witnesses, shaded: exists-unique is false. The negation is true in the two shaded columns — none, or more than one. Form A splits the two failure modes: not exists-unique is equivalent to (for all x , not $P(x)$) or (there exist y and z with $P(y)$ and $P(z)$ and y not equal to z), read as either nothing works or at least two different things do. Form B says no witness is ever alone: not exists-unique is equivalent to for all x , if $P(x)$

then there exists y with $P(y)$ and y not equal to x , read as every witness has a companion, vacuously true when there are none.

Counting witnesses makes the answer obvious: $\exists!$ picks out **exactly one** column of three, so its negation is the union of the other two. Both forms above were checked exhaustively against every predicate on universes of size 0 through 4.

1.5 Remark — The Two Failure Modes Are Genuinely Different

It is tempting to answer this problem with a single clause, but no single clause will do — the claim has *two independent* ways of failing, and a correct negation must admit both. “There is no such x ” is too weak (it misses the case of two solutions); “there are two such x ” is also too weak (it misses the empty case). This is exactly why the answer is a **disjunction**. The lesson generalises: when a statement is secretly a conjunction, its negation is secretly a case split, and forgetting one case is the commonest error in negating $\exists!$.

1.6 Remark — Form B: Why the Two Cases Can Be Merged

Form B, $\forall x [P(x) \Rightarrow \exists y (P(y) \wedge y \neq x)]$, looks like it only covers the “two or more” case, but it silently covers the empty case too — by **vacuous truth** (Problem 2.2.6). If nothing satisfies P , the antecedent $P(x)$ is false for every x , so the conditional holds trivially and Form B is true. That is precisely the behaviour we need. So the apparent two-case split of Form A is absorbed into one universal statement, at the cost of the reasoning being less transparent. Both are correct; Form A is clearer to read, Form B is more compact to manipulate.

1.7 Remark — Why $y \neq x$ and Not Just Two Variables

The clause $y \neq z$ in Form A is doing essential work. Without it, $\exists y \exists z (P(y) \wedge P(z))$ is satisfied by taking $y = z$ to be the *same* single witness — so it would say merely “at least one thing satisfies P ,” which is far too weak. Bound variables in logic are not assumed distinct; if you want distinctness you must **write it down**. This is the same care

needed in reading $\exists!$ itself, whose uniqueness clause is stated as $y = z$ rather than by any implicit convention.

1.8 Remark — $\exists!$ Is Not a Primitive Quantifier

Although $\exists!$ is written like $\forall$ and $\exists$, it is a *defined abbreviation*, not a third quantifier — which is why negating it required unpacking rather than a memorised flip rule. The definition also needs the equality symbol $=$, so $\exists!$ is only available in a language that has equality; the plain quantifiers $\forall, \exists$ do not. A second common definition, worth recognising, is

$$\exists!x, P(x) \cong \exists x [P(x) \wedge \forall y (P(y) \Rightarrow y = x)],$$

which packs existence and uniqueness into a single existential (“there is an x that works and is the only thing that works”). Negating *this* form directly gives Form B in three steps, which is where Form B comes from.

1.9 Remark — Where This Shows Up in Practice

Uniqueness claims are everywhere in mathematics, and knowing their negation tells you exactly what a counterexample must look like. “Every linear system $Ax = b$ has a unique solution” fails in two ways — **no** solution (inconsistent system) or **infinitely many** (free variables) — which is precisely the trichotomy above, and precisely why linear algebra courses treat those two failures as separate cases. Likewise “the equation has a unique root,” “the limit is unique,” “the identity element is unique”: to disprove any of them you must either exhibit *nothing*, or exhibit *two distinct* objects. The negation rule tells you which hunt to start.